

James Maa

E-mail: jamesmaa17@gmail.com • Phone: (650) 388-0628 • Website: jamesmaa.com • Current Location: San Francisco, CA
LinkedIn: linkedin.com/in/jamesmaa • Github: github.com/jamesmaa • Goodreads: goodreads.com/jamesmaa
Spoken languages: English (Native), Mandarin Chinese 中文 (Fluent), Japanese 日本語 (Proficient, JLPT N1), Spanish (Proficient)

SUMMARY

Senior Software Engineer with 8+ years of experience specializing in building highly scalable backend systems, cloud infrastructure, and CI/CD pipelines for large-scale platforms.

WORK EXPERIENCE

Career Break

Jan 2024 – Present

- Intentional career break to focus on personal growth while preparing for next high-impact engineering role
- Grew my open-source project, Yomitan (popup dictionary for language learning) from 10,000 to 100,000+ users within a year
 - Added support for multiple languages, including whole word scanning, language-specific settings and grammatical logic
 - Led a team of collaborators to prioritize removing user friction for non-technical users by adding onboarding and UX features

Discord

Aug 2020 – Jan 2024

Senior Software Engineer, Platform & Services

Remote + Hybrid @ San Francisco, CA

- Shipped multiple projects to improve the deployment system tied to our monolithic API server resulting in going from 2-3 deploys a week performed by staff engineers to 10+ continuous deployments a day, increasing throughput by 2000%, reducing deploy times by 90%, and shrinking the average changeset size by 80%
 - Engineered and built a database migration tooling and integrated it across all of our environments (dev, CI, staging, production), providing a smooth local dev UX working with migrations in Postgres and Cassandra and automating hundreds of dangerous and manual db operations in production, removing significant operational burden from the database team
 - Transitioned our monolithic API server from running on virtual machines in Salt to containers in Kubernetes
 - Rolled out a new deployment strategy onto Buildkite to automate the CI/CD process of the API servers, enabling developers to move safer and faster
 - Introduced deployment analysis that evaluates deployment health of deployments based on custom metrics, removing the need for a trained human operator to shepherd the deployment and saving engineers 30+ hours a week
- Migrated Discord's legacy asynchronous task system in Celery to GCP Pub/Sub by providing a easy-to-use developer interface for defining asynchronous tasks and automating the managing, scaling, and monitoring of the underlying infrastructure resources
- Drove team and company wide reliability efforts through new telemetry instrumentation, monitoring, dashboards, and processes
- Owned all the API-related infrastructure and platforms, including our Kubernetes clusters, Terraform, Cloudflare workers, Celery, Pub/Sub, Redis, and Google Cloud Platform, serving as the central infrastructure and DevOps team at Discord

Scratch Services (Fintech/Loan Servicing)

April 2018 – Dec 2019

Software Engineer, Backend & Infrastructure

In person @ San Francisco, CA

- Implemented, tested, and released concurrency in our core accounting infrastructure to increase the system concurrency level from 1 to 30 and expand our core system capacity by more than 1800%
- Delivered multiple mission-critical loan mechanic features like principal disbursements and forbearances on top of a custom-built temporal database with a transaction-log-inspired service handling out-of-order delivery of order-dependent computation, managing \$200M of debt
- Automated multiple manual business operation tasks that saved the company more than 30 hours a week of manual work including automated onboarding, loan sales, and adjustments

Reddit

April 2016 – April 2018

Software Engineer, Ads

In person @ San Francisco, CA

- Wrote the campaign management backend from scratch that is the source of truth for tens of thousands of ad campaigns resulting in a 350% increase in revenue from self-service advertisers
- Remodeled the payments and delivery data models to stomp out rampant data discrepancy issues in our original payment system. The new billing service successfully onboards hundreds of new advertisers and processes \$300k/week of charges
- Designed and wrote the inventory forecasting service that uses time series machine learning models to predict inventory and is used for pacing for all of Reddit's ad campaigns

Internships

Summers of 2013 – 2015

- 2013 @ Yahoo!: Wrote a data processing ETL to generate novel statistical insights for tens of thousands of online merchants
- 2014 @ Pinterest: Designed, trained, and productionized an adaptive logistical regression machine learning model that detects suspicious logins, improving recall by 16x and caught 42x more suspicious logins than the previous implementations
- 2015 @ Pinterest: Product Management Intern managing a team of 6 engineers shipping Pinterest's SDK to public release

EDUCATION & SKILLS

UC Berkeley

Skills

B.A. Computer Science Mastery: Google Cloud Platform, Kubernetes, Terraform, Cassandra, CI/CD, Buildkite, Pub/Sub

Aug 2012 - Dec 2015 Proficient: Networking & traffic, Cloudflare, SRE, Datadog, Observability

GPA: 3.7 Familiar with: AWS, PostgreSQL, Chef, Jenkins

Languages: Python (mastery), Node/Javascript (familiar), Ruby (familiar)